

Multiplication and division of algebraic fractions

Straight across for a product; turn the second one over for a quotient — and cancel before you multiply.

LESSON 118–119

2 × 40 MIN

MATEMATIKA 7, §53

S8 7.4–7.5

LESSON CLOCK

15'

20'

25'

20'

5'

Multiplying	15 min
Cancel first	20 min
Dividing	25 min
Powers, and all the excluded values	20 min
Homework	5 min

LEARNING OBJECTIVES

- Multiply algebraic fractions and cancel before multiplying out.
- Divide by multiplying by the reciprocal.
- Raise an algebraic fraction to a power.
- State every excluded value, including those from the divisor.

TEXTBOOK

National	pp. 350–357
Cambridge	Stage 8, pp. 75–78
Workbook	Exercise 7.5

01

Explanation

Multiplying

$$\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$$

No common denominator is needed — multiplication is the easy operation on fractions, not the hard one.

PRODUCT	STRAIGHT ACROSS	SIMPLIFIED
$\frac{2}{x} \cdot \frac{x}{3}$	$\frac{2x}{3x}$	$\frac{2}{3}$
$\frac{3a}{4} \cdot \frac{2}{9a^2}$	$\frac{6a}{36a^2}$	$\frac{1}{6a}$
$\frac{x+1}{x} \cdot \frac{x}{x+1}$	$\frac{x(x+1)}{x(x+1)}$	1

CONTRAST THIS WITH ADDING

Adding needed a common denominator and brackets; multiplying needs neither. Pupils often import the common denominator into a product out of habit — it is not wrong, just wasted work.

Cancel first

Factorise everything, cancel any factor of a numerator against any factor of a denominator, and only then multiply what is left.

STEP	WORKING
the product	$\frac{x^2 - 9}{x + 1} \cdot \frac{x + 1}{x - 3}$
factorise	$\frac{(x - 3)(x + 3)}{x + 1} \cdot \frac{x + 1}{x - 3}$
cancel $x + 1$ and $x - 3$	$x + 3$

MULTIPLYING OUT FIRST MAKES THE WORK FAR HARDER

Expanding $(x^2 - 9)(x + 1)$ produces a cubic that then has to be factorised again to cancel. Cancel first: it is the difference between one line and five.

Dividing

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \cdot \frac{d}{c} = \frac{ad}{bc}$$

QUOTIENT	AS A PRODUCT	ANSWER
$\frac{2}{x} \div \frac{4}{x^2}$	$\frac{2}{x} \cdot \frac{x^2}{4}$	$\frac{x}{2}$
$\frac{x^2 - 1}{x} \div \frac{x + 1}{x^2}$	$\frac{x^2 - 1}{x} \cdot \frac{x^2}{x + 1}$	$x(x - 1)$
$\frac{a}{b} \div a$	$\frac{a}{b} \cdot \frac{1}{a}$	$\frac{1}{b}$

INVERT THE DIVISOR, NEVER THE DIVIDEND

It is the **second** fraction that is turned over. Inverting the first gives the reciprocal of the right answer, and the mistake is invisible unless the answer is checked with a number.

Powers, and all the excluded values

$$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$$

EXPRESSION	ANSWER
$\left(\frac{2}{x}\right)^3$	$\frac{8}{x^3}$
$\left(\frac{3a}{b^2}\right)^2$	$\frac{9a^2}{b^4}$

In a division, three things must be non-zero: both original denominators, and the numerator of the divisor — because it becomes a denominator when the fraction is inverted.

DIVISION	EXCLUDED	WHY
$\frac{x^2 - 1}{x} \div \frac{x + 1}{x^2}$	$x \neq 0$	both denominators
$\frac{x^2 - 1}{x} \div \frac{x + 1}{x^2}$	$x \neq -1$	it becomes a denominator

THE HIDDEN EXCLUSION

$x = -1$ never appears as a denominator in the question as written, yet it is forbidden. Dividing by zero is what the rule is protecting against, and the zero is in the numerator of the divisor.

02 Worked examples

EXAMPLE 1 Simplify $\frac{x^2 - 9}{x + 1} \cdot \frac{x + 1}{x - 3}$.

① $x^2 - 9 = (x - 3)(x + 3)$

② Cancel $x + 1$ and $x - 3$.
Before multiplying anything out.

③ $= x + 3$
With $x \neq -1$ and $x \neq 3$.

ANSWER

$x + 3$

EXAMPLE 2 Simplify $\frac{x^2 - 1}{x} \div \frac{x + 1}{x^2}$.

① Invert the second fraction: $\frac{x^2 - 1}{x} \cdot \frac{x^2}{x + 1}$.

② $x^2 - 1 = (x - 1)(x + 1)$

③ Cancel $x + 1$, and x against x^2 .

④ $= x(x - 1)$
With $x \neq 0$ and $x \neq -1$.

ANSWER

$x(x - 1)$

EXAMPLE 3 Simplify $(\frac{3a}{b^2})^2$.

① The power applies to numerator and denominator.

② $(3a)^2 = 9a^2$

③ $(b^2)^2 = b^4$

④ $= \frac{9a^2}{b^4}$

ANSWER

$$\frac{9a^2}{b^4}$$

03 Interactive model

Set the same product twice — once told to expand first, once told to cancel first — and time both halves of the class; the cancelling half finishes before the other has factorised.

Cancel first, multiply second

$$\frac{x^2 - 9}{x + 1} \cdot \frac{x + 1}{x - 3}$$

① $x^2 - 9 = (x - 3)(x + 3)$

Factorise
before
anything
else.

② $\frac{(x - 3)(x + 3)}{x + 1} \cdot \frac{x + 1}{x - 3}$

③ $x + 3$

Cancel
 $x +$
 1
and
 $x -$
 3 ; x
 $\neq$
 -1 ,
 3 .

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Multiplication	Ko'paytirish	Умножение
Division	Bo'lish	Деление
Reciprocal	Teskari son	Обратная дробь
To invert	Ag'darib qo'yish	Перевернуть
Cross-cancelling	Krest-nakrest qisqartirish	Сокращение крест-накрест
Power of a fraction	Kasrning darajasi	Степень дроби
Divisor	Bo'luvchi	Делитель
Excluded value	Mumkin bo'lmagan qiymat	Недопустимое значение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

To multiply two fractions you:

need a common denominator

multiply across

invert the second

add the numerators

To divide by a fraction you:

invert the first

invert the second

invert both

invert neither

$\frac{2}{x} \div \frac{4}{x^2}$ equals:

$$\frac{x}{2}$$

$$\frac{8}{x^3}$$

$$\frac{1}{2x}$$

$$\frac{2}{x}$$

In a product you should cancel:

after multiplying out

before multiplying out

never

only numbers

(2 x)³ equals:

$$\frac{6}{x^3}$$

$$\frac{8}{x^3}$$

$$\frac{8}{3x}$$

$$\frac{2}{x^3}$$

In $\frac{a}{b} \div \frac{c}{d}$ which must be non-zero?

b

b and d

b , c and d

none

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

$$1. \quad \frac{2}{x} \cdot \frac{x}{3}$$

$$\text{ANSWER } \frac{2}{3}$$

$$2. \quad \frac{3a}{4} \cdot \frac{2}{9a^2}$$

$$\text{ANSWER } \frac{1}{6a}$$

$$3. \quad \frac{x+1}{x} \cdot \frac{x}{x+1}$$

$$\text{ANSWER } 1$$

$$4. \quad \frac{2}{x} \div \frac{4}{x^2}$$

$$\text{ANSWER } \frac{x}{2}$$

$$5. \quad \frac{a}{b} \div a$$

$$\text{ANSWER } \frac{1}{b}$$

$$6. \quad \left(\frac{2}{x}\right)^3$$

$$\text{ANSWER } \frac{8}{x^3}$$

$$7. \quad \left(\frac{3a}{b^2}\right)^2$$

$$\text{ANSWER } \frac{9a^2}{b^4}$$

Medium · the standard question

7 PROBLEMS

1. $\frac{x^2 - 9}{x + 1} \cdot \frac{x + 1}{x - 3}$

ANSWER $x + 3$

2. $\frac{x^2 - 1}{x} \div \frac{x + 1}{x^2}$

ANSWER $x(x - 1)$

3. $\frac{x}{x + 2} \cdot \frac{x^2 - 4}{x}$

ANSWER $x - 2$

4. $\frac{4}{x^2} \div \frac{2}{x}$

ANSWER $\frac{2}{x}$

5. $\frac{x^2 + 5x + 6}{x + 2} \cdot \frac{1}{x + 3}$

ANSWER 1

6. The excluded values of $\frac{x^2 - 1}{x} \div \frac{x + 1}{x^2}$

ANSWER $x \neq 0$ and $x \neq -1$

7. $\frac{a^2}{b} \cdot \frac{b^2}{a}$

ANSWER ab

Hard · stretch and reason

7 PROBLEMS

1. $\frac{x^2 - 4}{x^2 + 4x + 4} \cdot \frac{x + 2}{x - 2}$

ANSWER 1

2. $\frac{x^3 - 8}{x^2 - 4} \cdot \frac{x + 2}{1}$

ANSWER $x^2 + 2x + 4$

3. $\frac{a^2 - b^2}{a + b} \div \frac{a - b}{a}$

ANSWER a

4. $(\frac{x}{y})^2 \cdot \frac{y^3}{x}$

ANSWER xy

5. $\frac{x^2 - 9}{x^2 - 4} \div \frac{x + 3}{x + 2}$

ANSWER $\frac{x - 3}{x - 2}$

6. Why must the numerator of the divisor be non-zero?

ANSWER It becomes a denominator when the fraction is inverted

7. $\frac{1}{x} \div \frac{1}{x^2} \div x$

ANSWER 1

07 Homework

Homework — 5 tasks

Factorise and cancel before multiplying anything out.

1. Simplify $\frac{3}{x} \cdot \frac{x^2}{6}$ and $\frac{5a}{2} \cdot \frac{4}{15a^2}$.

2. Simplify $\frac{x^2 - 16}{x + 2} \cdot \frac{x + 2}{x - 4}$.

3. Simplify $\frac{x^2 - 4}{x} \div \frac{x + 2}{x^2}$.

4. Simplify $(\frac{4}{a})^2$ and $(\frac{2x}{y^3})^3$.

5. List every excluded value of $\frac{x + 1}{x - 2} \div \frac{x + 3}{x}$.